

Project Proposal

Title of the Project

Secure TCP-Based Proxy Tunneling System with Encryption and Traffic Analysis

1. Problem Statement and Motivation

In many real-world network environments such as corporate offices, universities, and public Wi-Fi networks, internet access is regulated through firewalls and filtering systems. While such controls are necessary for enforcing organizational policies and maintaining security, data transmitted across these networks remains vulnerable to interception, tampering, impersonation attacks, and denial-of-service attempts.

Unsecured communication channels can expose sensitive information through packet sniffing or man-in-the-middle attacks. Furthermore, users in restricted environments may require secure communication channels to access legitimate external resources.

The motivation of this project is to design and implement a secure proxy tunneling system that demonstrates how fundamental network security principles can be applied to protect data in transit. The project aims to provide a practical implementation of secure communication mechanisms along with detailed network traffic analysis.

2. Core Objectives of the Project

The primary objectives of this project are:

1. To design and implement a secure TCP-based proxy tunneling system.
2. To establish an encrypted communication channel between a client and a remote server.
3. To implement mutual authentication using a challenge–response mechanism.
4. To ensure confidentiality and integrity of transmitted data using cryptographic techniques.
5. To implement digital signatures to provide non-repudiation during session establishment.
6. To enforce access control through IP whitelisting.
7. To improve availability through rate limiting and connection management.

8. To analyze and evaluate network traffic using Wireshark to validate the effectiveness of the implemented security mechanisms.
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3. Overview of the Proposed System

The proposed system consists of two primary components:

- **Client Module**
- **Server Module**

The client acts as a local HTTP/HTTPS proxy that intercepts browser traffic. It encrypts outgoing requests and forwards them over a TCP connection to a remote trusted server.

The server decrypts incoming requests, forwards them to the intended destination web server, retrieves the response, encrypts it, and sends it back to the client. The client then decrypts the response and forwards it to the browser.

The communication between client and server forms a secure encrypted tunnel. Additional security enhancements such as mutual authentication, digital signatures, IP-based access control, and rate limiting are integrated to satisfy fundamental network security requirements.

4. Network Architecture and Communication Model

The system follows a client–server proxy architecture:

1. The web browser sends a request to the local proxy (client module).
2. The client encrypts the request using symmetric encryption.
3. The encrypted data is transmitted over TCP to the server.
4. The server authenticates the client and verifies digital signatures.
5. The server decrypts the request and forwards it to the target web server.
6. The response from the target server is encrypted and sent back to the client.
7. The client decrypts and forwards the response to the browser.

The architecture ensures secure tunneling of browser traffic over potentially untrusted networks.

5. Security Mechanisms to be Implemented

The project addresses the following fundamental network security principles:

Confidentiality

Implemented using AES encryption to protect data from unauthorized disclosure.

Integrity

Ensured using cryptographic message authentication (HMAC) and digital signature verification to detect tampering.

Authentication

Implemented using a mutual challenge–response mechanism between client and server.

Non-Repudiation

Achieved using Digital Signature during session establishment, ensuring that entities cannot deny their actions.

Availability

Enhanced using rate limiting and connection timeout mechanisms to mitigate basic denial-of-service attempts.

Access Control and Authorization

Implemented through IP whitelisting to restrict access based on predefined security policies.

6. Tools, Technologies, and Protocols

Programming Language:

- Python 3.x

Libraries and Modules:

- cryptography (Fernet for AES encryption)
- socket (TCP communication)
- requests (HTTP forwarding)
- threading (multi-client handling)

Protocols Used:

- TCP (Transport Layer)
- HTTP/HTTPS (Application Layer)

Network Analysis Tool:

- Wireshark for packet capture and traffic inspection.

7. Expected Outcomes and Evaluation Criteria

Expected Outcomes

- A functional secure proxy tunneling system.
- Demonstration of encrypted communication over TCP.
- Successful implementation of authentication, integrity, non-repudiation, availability, and access control mechanisms.
- Comparative analysis of encrypted vs non-encrypted traffic using Wireshark.
- Identification of system limitations and security trade-offs.

Evaluation Criteria

The project will be evaluated based on:

- Correct implementation of encryption and authentication mechanisms.
- Effectiveness of access control and availability features.
- Depth and clarity of network traffic analysis.